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Behavioral Responses of Bonnethead Sharks (*Sphyrna tiburo*) to Controlled Olfactory Stimulation

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Swimming behaviors of bonnethead sharks (*Sphyrna tiburo*) were observed in response to chemical stimulation. Large volume (10 ml) samples were introduced into the path of swimming sharks. In still water, tight turning and circling were observed while samples introduced in the presence of a water current elicited a swimming pattern of connected loops moving downcurrent. A headmount attached to the shark allowed small volume (0.5 ml) samples to be directed into the olfactory capsules. Stimulation in still water produced the turning and circling behaviors as observed for the large volume samples and stimulation in moving waters produced the same connected loops pattern. Other experiments confirmed that the small volume samples quickly dilute below threshold and therefore the behaviors observed in response to the 0.5 ml samples are swimming patterns released by the chemical signal but are presumed to be modulated by factors other than chemical cues. Additionally, it was demonstrated that bonnethead sharks can detect and respond to concentration differences between the left and right olfactory capsules. This means that these sharks are capable of tropotactic orientation in a chemical gradient.

INTRODUCTION

Orientation to food by means of chemical cues is wide-spread in nature and may be accomplished by several different behavioral

mechanisms (Bell and Tobin, 1982). In aquatic and marine environments, chemical stimuli are often quite persistent and chemosensory orientation may be particularly important. The ability of elasmobranchs to localize the source of chemical signals is well-documented and attempts to understand the neural and behavioral determinants of such chemosensory orientation suggest that both true gradient search and stimulus released rheotaxis are involved (Hodgson and Mathewson, 1971). The considerable separation between the right and left sides of the nose in large fish or in shark species like hammerheads and bonnetheads suggest that tropotaxis (instantaneous comparison of concentration at paired receptors) as well as klinotaxis (comparisons over time) is possible.

The behavioral mechanisms involved in chemosensory orientation typically are inferred from the patterns of movement of the animal relative to the presumed spatial and temporal features of an applied stimulus. On the basis of such information Hodgson and Mathewson (1971) concluded that the nurse shark (*Ginglymostoma cirratum*) reached the source of stimulation by gradient search while the lemon shark (*Negaprion brevirostris*) oriented against the current after encountering the chemical stimulus. Kleerekoper *et al.* (1975), also using nurse sharks, concluded on the basis of a detailed analysis of turning behavior that flowing water was necessary for precise localization of a stimulus source. In stagnant water only generalized localization occurred.

Analysis of swimming behavior is often based on observation over long periods of time during which the angles of many turns are averaged (Kleerekoper *et al.*, 1975). An attempt is then made to correlate the resulting "net" behaviors with presumed spatial and temporal changes in stimulus concentration. Although this averaged behavior is a reasonable measurement under conditions where the stimulus is subject to considerable mixing and dilution, much more precise information about individual turns as the animals seeks the source of a chemical is needed if one wishes to understand fully the behavioral mechanisms of gradient searching. This in turn requires that the spatial and temporal features of the stimulus be known.

This paper describes experiments conducted with a stimulus delivery system which permits precise control of temporal and spatial variations in concentration at the olfactory organs. It is possible to correlate observed swimming behaviors with specific stimulus condi-

tions and thus gain a more complete understanding of the physiological and behavioral mechanisms of chemosensory orientation.

MATERIALS AND METHODS

Bonnethead sharks (*Sphyrna tiburo*) were captured by gill net in the Gulf of Mexico near the Florida State University Marine Laboratory. Fish were immediately placed in 1.2 m diameter tanks and returned to the laboratory where they were transferred to 650 l, 1.8 m diameter observation pools supplied with continuously flowing sea water (6 l/min). Headmounts and stimulus delivery tubes were attached to the three sharks used in these experiments (50.8 cm; 66.0 cm; 48.3 cm TL) after a 12 hr recovery period.

A plastic headmount ($24 \times 8 \times 10$ mm) was attached to the dorsal surface of the shark's head with cyanoacrylate glue (Fig. 1). This

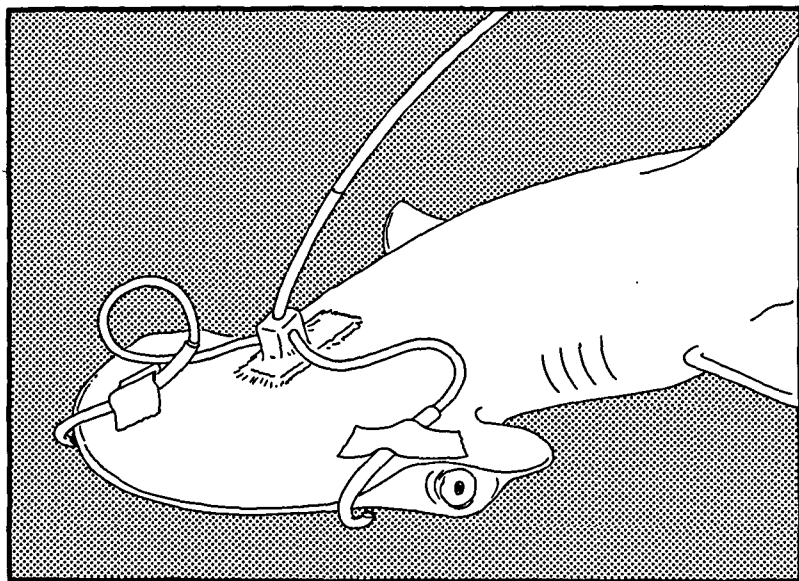


FIGURE 1 Bonnethead shark with a headmount glued to its dorsal surface. Two tubes from the stimulus reservoirs pass through the headmount and then into glass tubes which direct stimuli into the incurrent nares. Tape on the glass tubes was necessary to facilitate gluing them into position. Swimming movements were recorded from the video screen by noting the position of the tape marks.

acted as an anchor for the two polyethylene tubes (1.14 mm I.D.) which ran from the stimulus reservoirs, suspended above the tank, to the nostrils of the animal. The fish would not tolerate tubes placed directly into the incurrent nares. Consequently, glass tubes, attached to the polyethylene tubing on the headmount and bent so that their tips were immediately adjacent to but not touching the openings of the nostrils, were used to apply stimuli. The two stimulus reservoirs could be filled with 0.5 ml of stimulus via three-way valves. The upper ends of these reservoirs were connected by a Y tube, and a single piece of polyethylene tubing then passed through a swivel canula and ran to a syringe filled with sea water. A syringe driver was used to pump seawater through the system thus forcing the stimuli from the reservoir through the headmount and into the fish's nostrils. The delivery tubes between the stimulus reservoir and the fish were also filled with sea water so that during activation of the pump and before arrival of the stimulus in the nose, sea water was flowing through the olfactory capsules. The relatively long lengths of tubing between the reservoirs and the fish's nose provided a prestimulus period during which the effects of pumping water into the nostrils could be observed. However, it also resulted in a slight dilution of the leading and trailing edges of the stimulus pulse. Dye tests indicated that the stimulus reached a maximum concentration in the olfactory capsule of about 50% of the applied concentration within 7 sec after activating the syringe pump and was subsequently diluted below threshold within 10 sec. These dye tests also confirmed that there was no crossover of stimuli between the left and right nostrils. Thus, this application procedure insured a high degree of precision in temporal (time of arrival), concentration and spatial distribution of the delivered stimuli relative to the shark's olfactory receptor organs.

Stimulus solutions were made from whole homogenates of blue crabs (*Callinectes sapidus*), a natural prey of bonnetheads. Three small crabs (15 cm carapace width) were minced in a blender and thoroughly mixed with 2 liters of filtered sea water. This solution was filtered through a 0.5 mm mesh and 1400 ml of liquid was recovered and refrigerated. Dilutions were prepared for each experiment from this stock solution.

Swimming behaviors of the fish were recorded on video tape and then traced from the video monitor screen onto clear acetate sheets.

The animal's movements in response to either unilateral and bilateral stimulation of the olfactory system or to simulated gradients between the right and left nostrils were analyzed from these tracings. A turn, either toward or away from the stimulus, was defined as a deviation of 20 degrees or greater from the direction of travel just prior to stimulus arrival; a "no response" score was given to deviations of less than 20 degrees. Fish typically turned toward crab extract and consequently, turns toward the stimulus or the higher concentration of a simultaneously presented pair were scored as correct responses and turns away from the stimulus as incorrect.

RESULTS

To assess the effect of the headmount and stimulus delivery tubes on the swimming behavior of the fish, swimming speeds before and after the headmount was attached were compared for all three sharks. The mean swimming speed shortly after the sharks were transferred to the observation pool was 44 cm/sec while swimming against a current estimated to be 15 cm/sec during dye tests. A similar swimming speed was observed after attachment of the headmount. Two hours after handling, the swimming speeds slowed to 33 cm/sec for fish both with and without headmounts. Although these results indicate that disturbed sharks swim faster than undisturbed animals, the headmounts themselves did not appear to have any effect on swimming speed nor did they cause any unusual or erratic behavior.

After a minimum of 12 hr following capture, the responses of sharks to crab extract were observed prior to the attachment of the headmounts. When undisturbed sharks in a tank with no water flowing encountered a 10 ml bolus of crab extract poured into the tank they would stop their characteristic circling around the outside of the tank and go into a tight turning behavior remaining in the vicinity of the stimulation site. An example of this type of behavior is illustrated in Fig. 2A and was repeated for at least three trials for each of the three sharks.

Experiments where 10 ml aliquots of dye were introduced into a 15 cm/sec current flowing around the circumference of the tank indicated that the stimulus traveled around the tank as a more-or-

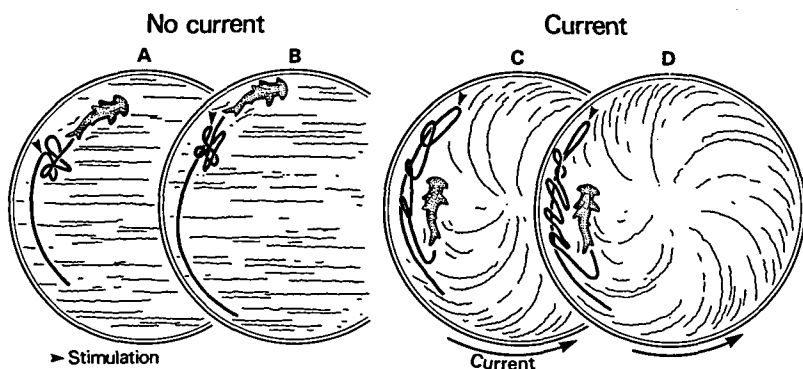


FIGURE 2 Swimming paths of bonnethead sharks recorded by an overhead video camera. A—Response to 10 ml sample of crab homogenate introduced in still water; B—Response to 0.5 ml sample delivered *via* head mount; C—Response to 10 ml sample delivered into 15 cm/sec water current; D—Response to 0.5 ml sample delivered *via* headmount in presence of 15 cm/sec water current.

less discrete bolus. Fish swimming against the current would reverse direction upon encountering a 10 ml sample of crab extract. As they moved downstream and presumably swam beyond the stimulus, the fish would again reverse direction and swim against the current. This behavior of reversing swimming direction was repeated as sharks appeared to follow the stimulus bolus around the edge of the tank. The paths of the fish then had the appearance of connected loops (Fig. 2C) and were observed for at least three presentations for each shark.

Three trials of small volume, bilateral stimulation (0.5 ml simultaneously into each nostril) of the three sharks circling in still water resulted in a tight turning behavior similar to that observed when sharks encountered a 10 ml bolus in still water (Fig. 2B). When the sharks were circling the tank in the presence of a current, a 0.5 ml bilateral stimulation elicited a reversal of swimming direction as when the 10 ml bolus was encountered (Fig. 2D). This looping behavior persisted through several cycles as the sharks moved down-current, even though studies indicated that a 0.5 ml sample introduced directly in the path but 15 cm ahead of a swimming shark elicited no observable response and thus was presumably diluted to below behavioral threshold levels very rapidly. This behavior resulted in the connected loop pattern of movement very similar to that exhi-

bited by sharks encountering larger volumes of stimuli. With bilateral stimulation the direction which the sharks turned was random (4 right, 6 left). Control applications of sea water resulted in no turns either for bilateral (0 turns in 2 trials) or unilateral (0 turns in 3 trials) presentations. These results indicate that small, brief stimuli can release locomotor patterns which long outlast the behavioral efficacy of the stimulus signals.

Experiments to measure the ability of the shark to detect a spatial concentration gradient were complicated by the shark's persistent behavior of closely following the outside wall while circling the tank. Unilateral stimulation of circling fish failed to elicit consistent turns to the side of stimulation. The shark would almost invariably turn toward the wall of the tank whether the stimulus was delivered to the left or right nostril. The sharks would, however, make occasional excursions across the center of the tank. During these periods unilateral stimulation resulted in turns toward the stimulated side or the side stimulated with the higher concentration (8 correct in 8 trials), followed by the characteristic tight turning behavior.

Although no attempt was made to determine the minimum detectable difference in concentration between the right and left nostrils, sharks did respond to simultaneously applied pairs of stimuli of full strength (100%) crab extract and half strength (50%) extract by turning toward the higher concentration in 5 out of 5 trials.

DISCUSSION

The most striking result of these experiments is the observation of swimming patterns, triggered by olfactory stimuli, which once released appeared to be modulated by factors other than chemical cues. The locomotor patterns evoked by small volume, short duration stimuli were similar to those associated with persistent, large volume stimulation. The looping pattern of swimming observed when an animal encounters a bolus of stimulus in moving water previously was thought to result from detection of a chemical gradient and a reversal in direction when the shark swims beyond the stimulus into unscented water (Kleerekoper, 1982). In this way, the animal would presumably detect the stimulus while swimming

against the current, then pass out of the stimulus, turn and loop back into the stimulus as it moves around the tank with the current. In the experiments described above, however, where headmounts were used to deliver very brief pulses of stimulus, additional or alternative mechanisms must be involved because the stimulus is rapidly diluted below threshold. Fish exhibiting turning behavior in still water after stimulation with small volumes of stimuli also appear to be displaying a chemically released searching behavior which functions to keep the animal in the vicinity of the initial stimulation site. Similar observations have been made for the channel catfish (*Ictalurus punctatus*) and the brown bullhead (*I. nebulosus*; Johnsen, unpublished observations).

The behavior of the fish stimulated in a current is more complicated in that a directional bias is added to the turning behavior. Rather than exhibiting tight turning at the site of stimulation, the animal searches down-current through a looping swimming pattern during which chemical cues appear to play little or no role beyond releasing the swimming pattern.

Observation of apparently oriented movements which are thought to no longer be under the direct influence of chemical cues have been described in moths (Marsh *et al.*, 1978), the American cockroach (Tobin, 1981) and coho salmon (Johnsen, 1982). In both the cockroach and salmon an internal mechanism has been suggested for controlling the turning behavior at the interface between the scented and unscented environment. Tobin (1981) suggested that these internally generated turning behaviors may serve to maintain the oriented course after temporary disruptions in the odor trail. In the case of sharks, these released turning behaviors may allow the animal to average the chemical information available and then integrate it with other cues. If the animal is not rigidly fixed to physiological rules governing its movements, such as always turning in the direction of highest concentration of a natural feeding stimulant, it would be able to optimize its path in a trail of chemical stimuli containing deviations in its gradient characteristics. Thus, a shark averaging chemical gradient information and integrating it with other cues such as water currents could maximize its searching efficiency.

Although orientation of sharks to chemical gradients is often attributed to sequential comparisons of concentration over time

(klinotaxis), our experiments using controlled applications of pairs of stimuli to the right and left olfactory capsules indicate that bonnethead sharks are capable of tropotaxis as well. Additional experiments will be needed to determine how small a concentration difference between the left and right sides of the nose can be detected. Such studies should provide critical information concerning the importance of various mechanisms of chemosensory orientation in these animals.

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